

Secure Exit

CO3-AT3-Assignment

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SIMATS Online Programming Lab

JAVA PROGRAMMING

 Akileshwaran A 192511354RD ▼

QUESTIONS

Q1Banking Transaction System –
Exception Handling

10 marks

Q2Hospital Patient Registration –
User-Defined Exception

10 marks

Q3E-Commerce Order Processing
– Built-in Exceptions

10 marks

Q4Hospital Emergency System –
Thread Priority

10 marks

Q5Online Food Delivery – Multiple
Threads and Execution Order

10 marks

5/5 answered

Q3 E-Commerce Order Processing – Built-in Exceptions

10 marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 `ArrayIndexOutOfBoundsException` handled Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 `InputMismatchException` handled Invalid input

Your Code

```

1 import java.util.*;
2 public class Main{
3     public static void main(String args[]){
4         Scanner sc=new Scanner (System.in);
5         String[]products={"Laptop", "phone", "Headphones"};
6         double[] prices={50000,20000,3000};
7         try{
8             int index=sc.nextInt();
9             int quantity=sc.nextInt();

```

```
10         if(quantity<=0){
11             throw new IllegalArgumentException("Invalid quantity.");
12         }
13         double total=prices[index]*quantity;
14         System.out.println("total Cost="+ total);
15     }
16     catch(ArrayIndexOutOfBoundsException e){
17         System.out.println("ArrayIndexOutOfBoundsException handled.")
18     }
19     catch(InputMismatchException e){
20         System.out.println("InputMismatchedException handled.");
21     }
22     catch(IllegalArgumentException e){
23         System.out.println("Invalid quantity");
24     }
25 }
26 }
```



Input

abc 2

Output

InputMismatchedException handled.

✓ Submitted

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